

Fourth Semester B. Tech. (Electronics and Communication Engineering / Electronics Design Technology) Examination

INTRODUCTION TO ELECTROMAGNETIC THEORY

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Assume suitable data wherever necessary.
- (3) Given Constants : $\epsilon_0 = 8.854 \times 10^{-12}$ F/m and $\mu_0 = 4\pi \times 10^{-7}$ H/m.
- (4) **Bold letters represents vector.**

1. Solve any **Six** questions :

- (A) A certain radiating antenna has radiation field $2\rho \cos 2\phi - \rho \sin 2\phi$. Find the radial part of field. 2(CO1)
- (B) Given three point A(2, 3, 2), B(-1, 0, 5) and C(5, -2, 1). Specify :
 - (i) A vector extending from origin to point A.
 - (ii) Give the vector extending from the origin to the midpoint of line AB. 2(CO1)
- (C) Find the total charge contained in the cylindrical volume defined by $\rho \leq 2$, $0 \leq z \leq 3$, if the surface charge volume $\rho_v = 20 \rho z$ mC/m³. 2(CO2)
- (D) Give the significance of Divergence, gradient and curl with suitable example. 2(CO3)
- (E) Derive an expression for magnetic field intensity due to an infinite straight conductor carrying current I in the positive Z-direction. 2(CO2)
- (F) State maxwell equation in integral form for time varying field. 2(CO4)
- (G) Explain the term total reflection and total transmission. Under what circumstances do they take place ? 2(CO3)
- (H) Calculate **E** at M(3, -4, 2) in free space caused by charge Q = 2 C at P(0, 0, 0). 2(CO1)

2. (A) A thick Spherical Shell centered at the origin extends between $R = 2$ cm to $R = 3$ cm. If the volume charge density is $\rho_v = 3R \times 10^{-4} \text{ C/m}^3$, find the total charge contained in shell. 3(CO3)
- (B) Given $\mathbf{D} = 2xy \mathbf{a}_x + x^2 \mathbf{a}_y \text{ C/m}^2$ parallelopiped dimensions are $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$, verify either side of Gauss divergence theorem. 4(CO3)
- (C) Given $V = 80r^{0.6} \text{ V}$, assuming free space, find the value of $\mathbf{E}$. 3(CO3)
3. (A) Calculate the value of vector current density :
- (a) In Cartesian coordinates at $\mathbf{P}_A(2, 3, 4)$ if $\mathbf{H} = x^2z \mathbf{a}_y - y^2x \mathbf{a}_z$;
- (a) In cylindrical coordinate at $\mathbf{P}_B(1.5, 90^\circ, 0.5)$ if $\mathbf{H} = \frac{2}{\rho} (\cos 0.2\phi) \mathbf{a}_\rho$. 4(CO3)
- (B) A dipole moment $\mathbf{p} = 6 \mathbf{a}_z \text{ n C}\cdot\text{m}$ is located at the origin in free space. Find V at $\mathbf{P}(r = 4, \theta = 20^\circ, \phi = 0^\circ)$. Further Evaluate $\mathbf{E}$ at $\mathbf{P}$. 3(CO2,3)
- (C) A current filament carrying 15 A in the $\mathbf{a}_z$ direction lies along the entire z axis. Find H in rectangular coordinates at $\mathbf{p}(20^{1/2}, 0, 4)$. 3(CO3)
4. (A) In cylindrical Coordinate system, the Magnetic field intensity is given by $H = (2\rho - \rho^2) \mathbf{a}_\phi \text{ A/m}$ for $0 \leq \rho \leq 1$. What is the total current density passing through the surface $Z = 0$, $0 \leq \rho \leq 1$ in $\mathbf{a}_z$ direction ? Verify the same with Stokes' theorem. 4(CO4,2)
- (B) Apply Faraday's Law to closed path and obtain Maxwell's Equation. 3(CO4)
- (C) Plane $z=0$ and $z=4$ carry current $\mathbf{K} = -10 \mathbf{a}_x \text{ A/m}$ and $\mathbf{K} = 10 \mathbf{a}_x \text{ A/m}$ respectively. Determine $\mathbf{H}$ at $(1, 1, 1)$ and $(0, -3, 10)$. 3(CO3)
5. (A) What is scalar magnetic potential ? Give its significance and comment on its multiple values using suitable example. 5(CO4)
- (B) Determine $\gamma, \alpha, \beta, v, \lambda$ for uniform plane wave propagating in perfect dielectric. Where symbols have usual meaning. 5(CO4)

6. Write short notes on (any **Two**) :—

- (i) Biot-Savart Law statement and explanation.
 - (ii) Isotropic and homogeneous medium.
 - (iii) Write differential form of Maxwell equation.
 - (iv) Magnetic flux, Magnetic flux Density and Magnetic vector potential.
- 8(CO2,3,4)

